

Comparing the Impact of Professional and Automatic Closed Captions on Video-Watching Experience

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Figure 1: (a) Example scene with professional CC, (b) example scene with automatic CC

ABSTRACT

Closed captions (CC) have gained popularity in recent years. While professional subtitles are commonly used, there has been a trend toward using automatic speech recognition technology to generate closed captions. However, there is limited research on the impact of these subtitles on viewers. To address this gap, we conducted an experiment to investigate how professional and automatic CC influence people's video-watching experiences. We developed to measure viewers' narrative engagement. We found that people showed better narrative understanding with the professional compared to the automatic one, while no significant difference in their emotional engagement and narrative presence. These findings suggest that automatic CC has the potential to be a rival professional one, but more research is needed to improve its effectiveness. Additionally, it provides insight into the motivations of viewers without

hearing impairments or language translation needs, who use CC to enhance their video-watching experiences.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in HCI**;
Empirical studies in interaction design.

KEYWORDS

closed captions, automatic speech recognition, narrative engagement, video-watching experience

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1 INTRODUCTION AND BACKGROUND

Closed captions, a form of subtitles that provide a textual representation of the audio in a media file, have gained increasing popularity in recent years [25]. This trend is exemplified by the success of the closed captions for the Netflix series, *Stranger Things* season 4,

which drew attention for its use of unique and descriptive captions that enhanced the viewing experience. Thanks to the widespread availability of streaming services and advances in sound recognition technology, closed captions are now easily accessible to many people. Streaming services such as Netflix, Hulu, and DisneyPlus provide professionally-generated closed captions that are generated by professional subtitle teams, referred to as “professional CC” (Figure 1(a)) [24]. On the other hand, video platforms such as YouTube use automatic speech recognition technology to generate closed captions referred to as “automatic CC” (Figure 1(b)) [26].

Closed captions are primarily utilized as a tool for accessibility for deaf and hard-of-hearing viewers [7]. However, according to World Wide Web Consortium (W3C), an international community that develops Web standards, automatic closed captions are not sufficient to provide accessibility, although they can serve as a starting point [11]. Given that a significant portion of viewers who do not have hearing impairments or need language translation also use closed captions these days [25], we aimed to compare how these two types of closed captions influence viewers watching narrative videos. To the best of our knowledge, the primary motivation for using closed captions and their effect on viewers have not been explored in depth.

1.1 Closed Captions

Compared to traditional subtitles that primarily translate the spoken dialogue of actors, closed captions also include non-speech elements such as sound effects, background music, or background sound [14]. These elements provide additional context and aid in viewers’ recognition of the mood of the scene. For instance, the closed captions of *Stranger Things season 4* became widespread due to their ability to effectively convey the atmosphere of the scene, such as the sounds of tentacles squelching wetly, thus creating an immersive viewing experience [22]. Other non-speech elements include the speaking style (e.g. whispering or shouting) and the identification of speakers when the speaker is off-screen or when there are multiple speakers, providing clarity as well as insight into the actors’ emotional states and nuances of action.

In February 2014, Federal Communications Commission (FCC) officially announced high-quality closed captions are required for television [5]. Quality standards of closed captions include (1) accurate alignment of all audio parts to the greatest extent possible, (2) synchronous with corresponding sounds and displayed to be easily read, (3) complete from the start to the end of the content, and (4) properly placed on the screen and not obscuring scenes. While professional closed captions adhere to this standard, automatic closed captions have several errors, including being out of sync with the audio and presented at a slight delay [6, 12, 23]. Furthermore, there are differences in display style between professional and automatic closed captions, with professional captions being displayed one line at a time and automatic captions being displayed one word at a time. [9]. Despite these issues, automatic closed captions have a considerable advantage in terms of the efficiency of the generation process and have therefore been the subject of research aimed at overcoming their drawbacks. However, these studies have only focused on improving accessibility for deaf or hard-of-hearing individuals [2, 10, 18]. In this paper, we investigate how closed

captions can enhance the viewing experience including for those without hearing impairments. By examining the benefits of closed captions beyond accessibility, we can develop design guidelines for automatic closed captions, improving their accuracy and quality.

1.2 Impact of Subtitle Types on Viewer Video-Watching

Previous research has recognized the importance of understanding the effects of subtitles on video watching. For example, Jian-xia et al. compared on-screen subtitles, pop-up subtitles, and no subtitles in a multimedia learning environment and found that learning outcomes varied depending on the subtitle presentation type, despite no differences in cognitive load [13]. Rajendran et al. studied the effects of text chunking on subtitles and found that subtitles presented word by word may be more challenging for viewers to process than subtitles chunked by phrases or lines [21]. In the HCI field, some studies have proposed interactive subtitle designs and evaluated the effects on watching videos. Brown et al. introduced dynamic subtitles that are positioned differently according to the content of the video [3]. They found that customizing subtitles’ appearance and position can improve the user experience. Kurzhals et al. proposed speaker-following subtitles, which move with the speakers in the scene, and found that this can reduce eye movements and improve viewers’ comprehension of the video content [16]. However, these studies are only limited to learning effects or showed that changing the position of subtitles can reduce eye strain and improve the viewer experience. Notably, these findings seem to contradict the growing trend of using subtitles even when accessibility support is not required. As such, investigating the reasons why people use subtitles or their engagement would be valuable in addressing the existing research gap in the HCI community.

1.3 Measuring Viewer Recognition and Engagement

There have been few studies that have investigated how to measure the extent that viewers recognize narratives or engage with videos. Perego et al. explored the impact of dubbed and subtitled content on people’s cognitive, evaluative, and visual reception [19]. The study used recognition tests consisting of a general comprehension test, a dialogue recognition test, a face-name association test, a visual scene recognition test, and a film evaluation test. Their research found that the cognitive process of subtitle materials and dubbed materials remains to a similar degree of effectiveness. Busselle and Bilandzic defined narrative engagement as a key measure of enjoyment and immersion in a narrative, which has been shown to have four dimensions: attentional focus, narrative understanding, emotional engagement, and narrative presence [4]. They found that the measure was able to differentiate between high and low levels of narrative engagement and was sensitive to manipulations of narrative elements that are known to affect engagement. Benini et al. examined the effect of shot scale on narrative engagement using this framework and found that more frequent close shots can increase engagement, while a higher frequency of long shots may negatively impact viewers’ focus on the plot [1]. Overall, the literature suggests that people’s recognition and engagement while watching videos are influenced by the design elements of the videos.

However, there has been limited research on how subtitle design specifically impacts audience narrative engagement.

In summary, previous work has presented possible influence factors of subtitles and shown various methods to measure their effects in video watching. Based on their findings, this study measures viewers' understanding and engagement of narratives to identify the different effects of professional CC and automatic CC. In the study, we also controlled the video stimuli by using three video clips with varying shot sizes, scales, and complexity as the previous research has shown that the video components influence viewers' watching experience in various ways [1, 15, 17, 19]. To our knowledge, this is the first study to compare the effects of these two types of captions on viewers' video watching. Our findings provide insight for researchers and practitioners in the field of video accessibility, interactive subtitle design, and highlight the potential for automatic closed captions to rival professional captions with further development.

2 METHODS

We conducted a between-subjects design in which participants were assigned to either professional CC condition or automatic CC condition. Three video clips were selected from YouTube that met the following inclusion criteria: (1) standard American English, (2) containing dialogue, (3) having both professional and automatically generated closed captions. We observed videos and chose three clips based on the speed of the videos. The speed of the video is manually coded as the average duration time of scenes, the number of scenes per 1 minute, and the number of words per 1 minute (Table 1).

To measure viewers' narrative understanding levels, we designed recognition tests for each video based on [19]. These tests were composed of three parts: 1) general comprehensive multiple-choice questions that measured participants' understanding of the plot of the video, 2) face-name association questions that required participants to correctly match characters with their names, and 3) visual-scene recognition questions that tested participants' ability to identify screenshots of scenes from the video stimuli. We also asked participants to rate their video-watching experience on a five-point Likert scale. Those questionnaires were developed to assess participants' emotional engagement and narrative presence. They were modified from the items of the narrative engagement survey developed by [4]. The emotional engagement questions included items focusing on how participants feel about the character's emotions (i.e. *Empathy*) and how participants feel for the characters (i.e. *Sympathy*). For the narrative presence, questions were adjusted focusing on how participants feel about transiting from reality to stories (i.e. *Narrative Presence*) and how participants feel the stories absorb their attention (i.e. *Narrative Involvement*).

To control for potential language effects on the results of the experiment, we recruited native English-speaking participants who were 18 years or older and had good vision. Participants were recruited through SONA, the university's online recruitment system, and received course credit as compensation for their participation. All aspects of the experiment were approved by the university IRB. A total of 89 participants were recruited, with 78 (35 female, all in the 18-24 age group) giving consent to participate in the experiment.

After screening out data from participants who did not complete the survey or spent less than 20 minutes or more than 2 hours on the experiment, our final sample included 52 participants (27 in the professional CC condition, 25 in the automatic CC condition). To gauge their familiarity with subtitles and video watching, we asked about their past experiences, and 17 participants indicated that they always use subtitles, 12 said they do so most of the time, 10 said they do so half of the time, and 9 said they rarely use subtitles.

During the experiment, participants were given a URL link to the online experiment portal through the SONA system. They were told the expected length of the experiment and asked to stay in an undisturbed environment during the experiment. Then, they had to complete general questionnaires to provide demographic information and record their previous engagement with video content and captions. The video stimuli were presented next, and finally, they got a questionnaire to measure the effects of either professional CC or automatic CC. Participants finished three video and questionnaire sessions in total.

3 RESULTS

We utilized the statistical software R studio [20] to obtain descriptive statistics and conducted *t*-tests to compare the professional CC group and the automatic CC group. The dependent variables we focused on were the accuracy rates of the recognition tests and the Likert scores of the surveys for emotional engagement and narrative presence. To ensure that the recognition tests were appropriately challenging and able to differentiate between high and low performers, we conducted a difficulty and discrimination analysis using Item Response Theory [8]. This mathematical model is commonly used in educational contexts to analyze assessment items. We also analyzed answers of the open-end questionnaires to understand the motivation of using closed captions.

3.1 Narrative Understanding

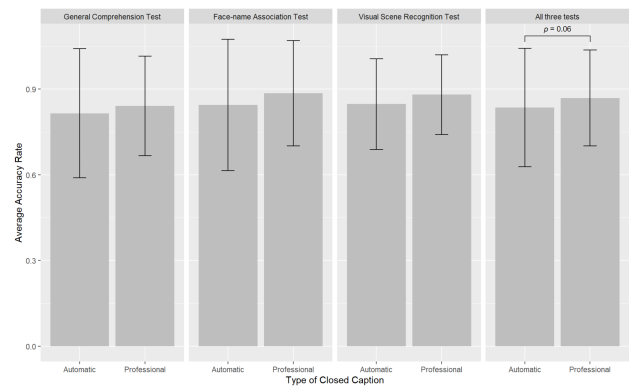


Figure 2: The average accuracy rate

From the recognition tests, we found that overall accuracy rates within the professional CC group ($Mean = 0.87$, $SD = 0.17$) were significantly higher than those within the automatic group ($Mean = 0.84$, $SD = 0.21$); $t(457.8) = 1.90$, $p = 0.06$. The individual sections of the recognition test did not differ significantly between

Table 1: Video Stimuli Speed

Video stimuli label	Fast Clip	Moderate Clip	Slow Clip
Film name	<i>Trouble Is My Business (2018)</i>	<i>Happily Ever After (2016)</i>	<i>The Secret Garden (2020)</i>
Genre	Romance, Crime, Thriller, Mystery, Adventure, Noir	Romance, Comedy, Drama	Children, Fantasy, Drama
Clip length	7'30"	5'48"	5'58"
Number of scenes / min	22.67	15.69	12.59
Average duration time of scenes (sec)	0.0578	0.0848	0.0943
Number of words / min	Automatic CC: 114.6, Professional CC: 132.7	Automatic CC: 111.3, Professional CC: 114.3	Automatic CC: 103.5, Professional CC: 113.8

the professional CC group and the automatic CC group. (General comprehension: $t(149.28) = 0.78$, Face-name association: $t(150.85) = 1.23$, Video scene recognition: $t(153.57) = 1.38$)

The item characteristic curves in Figure 3 illustrate the level of difficulty and the ability to distinguish participants in recognition tests with three videos. The average difficulty score for all questions was -129.62 ($SD = 343.34$), with scores for the slow, moderate, and fast videos being 3.34 ($SD = 16.37$), -78.00 ($SD = 280.99$), and -340.67 ($SD = 496.53$), respectively. These results indicate that the overall difficulty of the recognition tests was low, while the questions for the slow video were relatively difficult and the questions for the fast and medium videos were easier. The average discrimination score for all questions was 1.74 ($SD = 3.36$), with scores for the slow, moderate, and fast videos being 0.88 ($SD = 1.02$), 2.67 ($SD = 6.26$), and 1.75 ($SD = 4.79$), respectively. This suggests that the recognition tests were effective in distinguishing between participants with high and low levels of narrative understanding overall, with the questions for the moderate video demonstrating the best ability to discriminate between high and low-understanding participants.

According to prior research, presentation types of subtitles would affect viewers' understanding of video narratives [13]. Supporting these findings, the accuracy rates in recognition tests indicate that professional CC may aid viewers in comprehending the finer details of video plots, as it ensures that relevant dialogues are displayed at appropriate time points. By presenting professional closed captions sentence-by-sentence, participants would read the whole sentence and have concrete scripts in mind hearing the dialogue. This presentation method may also facilitate improved recall of character names, which are often mentioned in conversation or presented within brackets during speech. In contrast, automatic CC that displays words individually can require viewers to piece together the script themselves, potentially diverting their attention from the visual scenes in the video [21]. Errors, delays, and the absence of non-speech elements in automatic CC would also impede viewers from perceiving visual images. These additional efforts may distract people from the video content, making them unable to catch up with the storytelling flow of the video content.

3.2 Emotional Engagement and Narrative Presence

We also investigated Empathy (E1), Sympathy (E2), Narrative presence (N1), and Narrative involvement (N2). Figure 4 presents the average Likert point between the automatic closed caption group

and the professional closed caption group for each emotional engagement and narrative presence survey question. There were no significant differences between the two groups in all questionnaires from t -test. (Empathy (E1): $t(149.84) = 0.27$, Sympathy (E2): $t(150.96) = -1.02$, Narrative presence (N1): $t(153.92) = 0.91$, and Narrative involvement (N2): $t(153.35) = -1.44$. All questionnaires: $t(609.77) = -0.73$.) Our results show that professional and automatic closed captions have comparable effects on video viewers' emotional engagement and narrative presence. This result is encouraging as it indicates that even though automatic closed captioning temporarily has limitations such as higher error rate and delay time, it has the potential to work comparably as a complementary of professional closed captioning.

3.3 Previous experience on Closed Captions

From the survey, we found that 75% of the participants indicated they would turn on subtitles more than half the time when watching videos and 34 participants responded that they have used closed captions before. To better understand the motivation behind the usage of CC, we added open-ended questions in our survey asking participants about their reasons for using subtitles. Subsequently, we manually coded and categorized the responses through discussions. The results contain more than one type of category, the total number of the selected results is 52. By categorizing participants' responses, we found the majority (35 participants) use subtitles to more readily follow the plot and comprehend what the performers are saying, as shown in Table 2. Other reasons given included a personal preference for reading, difficulty hearing the audio of the video, and a desire to not disturb others.

The survey results indicate that participants widely used and valued subtitles for their ability to enhance understanding and engagement with video content. Given that the participants' age was between 18 to 24, it is in line with the previous survey [25]. The results also suggest that there are a variety of motivations behind the use of subtitles, highlighting the importance of providing subtitles as an accessibility feature for a diverse range of users. Many participants expressed that they use subtitles for reassurance and improved comprehension, with one participant stating, "sometimes I can't hear exactly what the people are saying and the captions help me get every part of the script." This suggests that visual and auditory images alone may not be sufficient for viewers to fully understand the content, particularly in cases where the video quality or sound settings are not optimal. Furthermore, some participants reported using subtitles due to environmental constraints, such as

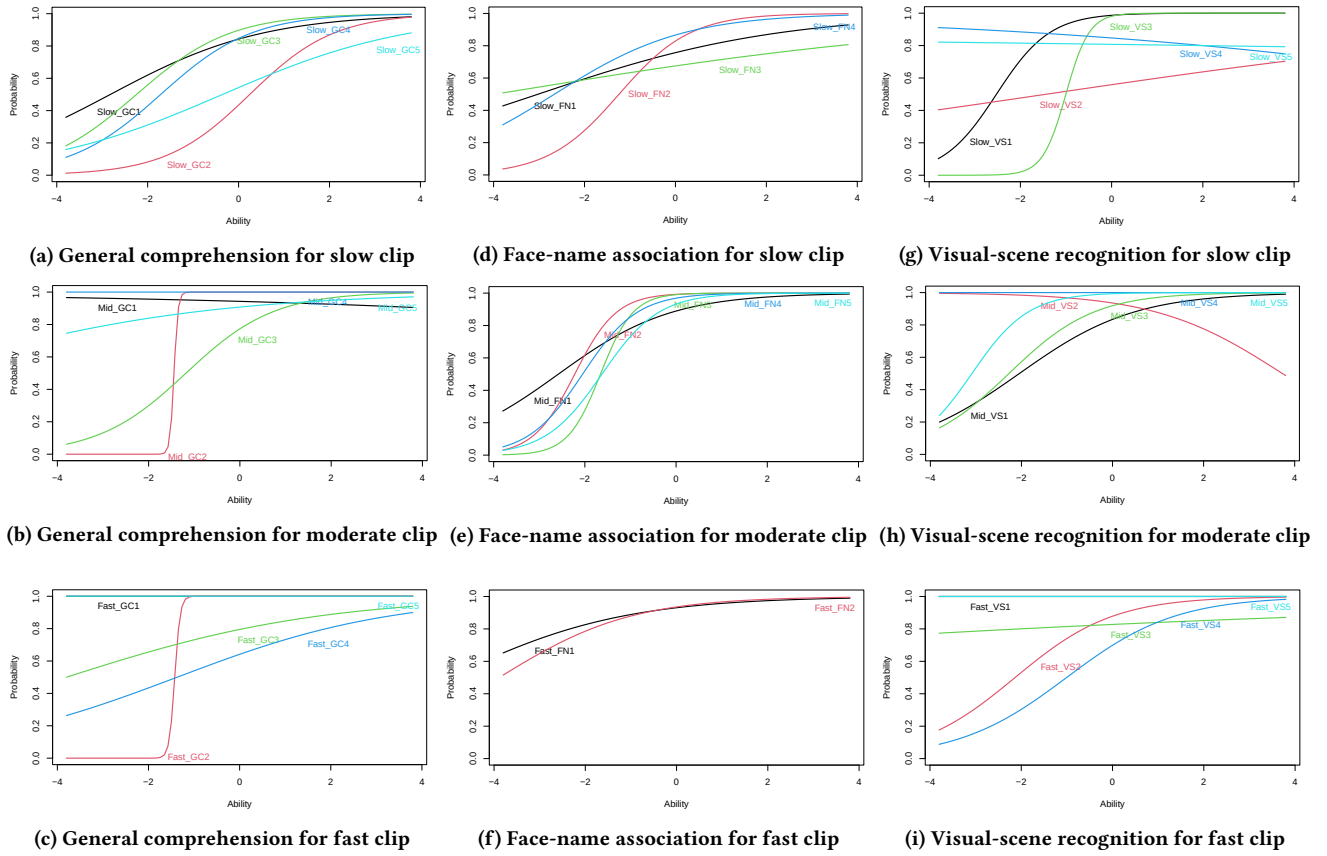


Figure 3: The item characteristic curves (x-axis: participants' overall narrative understanding level, y-axis: the probability of correctly answering a test question)

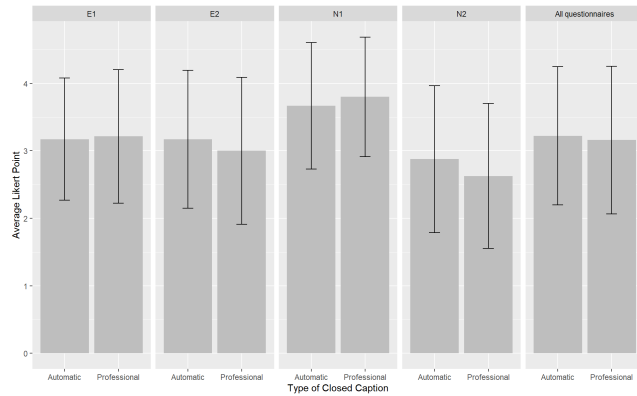


Figure 4: Average Likert Point Automatic CC and Professional CC Groups

surrounding noise. One participant noted, "if I'm eating or doing something loud, (subtitles) help to see what the video is saying." Interestingly, a few participants stated that they "don't want to keep the volume very high" not to disturb others around, suggesting that

subtitles would improve the personalized viewing experience in shared spaces where multiple users may be engaged in different activities.

4 DISCUSSION AND CONCLUSION

Our study found that professional CC improved viewers' understanding of narrative videos more than automatic CC, but there was no significant difference between the two types of CC in terms of emotional engagement or narrative presence. These findings suggest that while automatic closed captions may not yet be as effective as professional CC in facilitating viewer understanding of video storytelling, they have the potential to rival professional CC with further improvements. Specifically, improving weaknesses of automatic CC, such as display methods or detecting non-speech elements of video content, could aid viewers in understanding plot details. With the development of automatic speech recognition technology, it seems promising that we can save more human resources in writing closed captions and make the production process of closed captions more cost-effective.

However, there are several limitations to consider in interpreting the results of this study. First, the study was conducted online and remotely, meaning that it was difficult to control experiment

Table 2: Previous experiences on using subtitles

How often do you turn on subtitles?	# of Participants
Always	17
Most of the time	12
Half of the time	10
Rarely (only 1-2 times)	9
None	4
What types of subtitles have you used?	# of Participants
Closed captions (professional & automated)	20
Traditional subtitles	14
Both	14
No Answer	4
Motivations on turning on subtitles	# of Participants
Easy to understand and track the plot	35
Preference on reading	4
Hard to hear the audio of the video sometimes	17
Do not want to disturb others around	2

variables such as screen display, as the size and brightness of the display can affect participant attention and comprehension. Additionally, researchers could not guarantee that participants were undisturbed during the experiment or that they did not pause the video at any point. This could potentially impact the accuracy of the data collected. Second, the sample in this study was limited to college students, meaning that the results may not be generalizable to other demographic groups with different ages or educational backgrounds.

Still, the results of this study contribute to our understanding of the use of closed captions. The survey provides valuable insights into the motivations behind the use of subtitles and the ways in which they enhance viewers' understanding and engagement with video content. One interesting point to consider is that a majority of the participants in this study were already active users of both professional and automatic CC. Considering that several participants mentioned surrounding noises and video sounds as the reason for using subtitles, it may be worthwhile to explore the possibility of designing interactive closed captions that are activated based on surrounding noise levels or the volume of the video content. Additionally, given that one-third of participants mentioned using subtitles to improve their comprehension of the video content, it may be useful to consider designing a subtitle interface that includes summaries or bullet points to aid in understanding.

Future research could also include the analysis of eye-movement data to gain a deeper understanding of viewer behavior. This would require a more controlled experimental setup with the use of eye-tracking technology. Additionally, qualitative studies such as in-depth interviews or focus groups could provide further insights into viewer experiences with closed captions. Finally, it may be interesting to explore the potential of interactive closed captions, which could take advantage of viewer attention patterns to enhance engagement and potentially create new video-watching experiences.

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